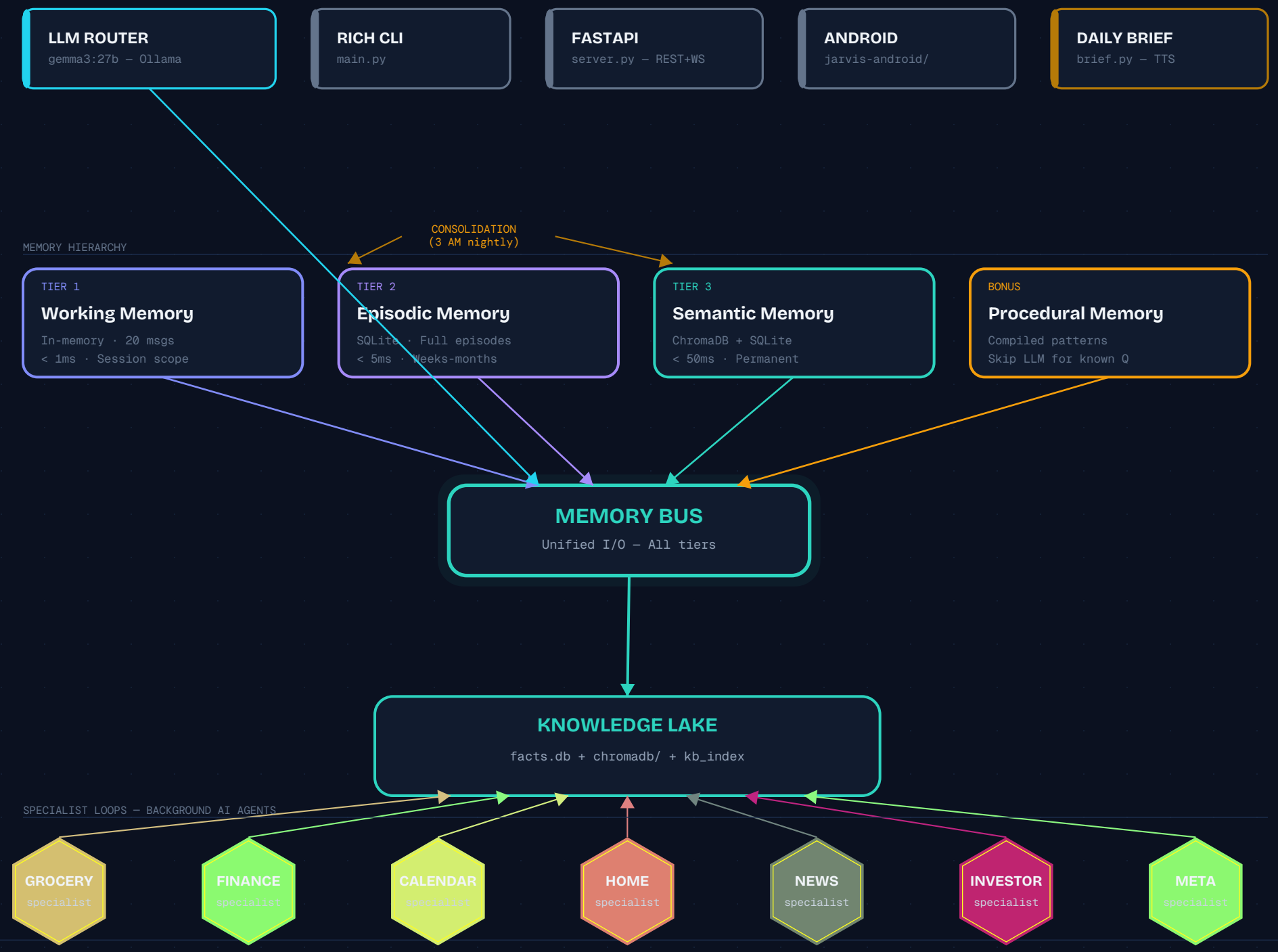
SYSTEM OVERVIEW

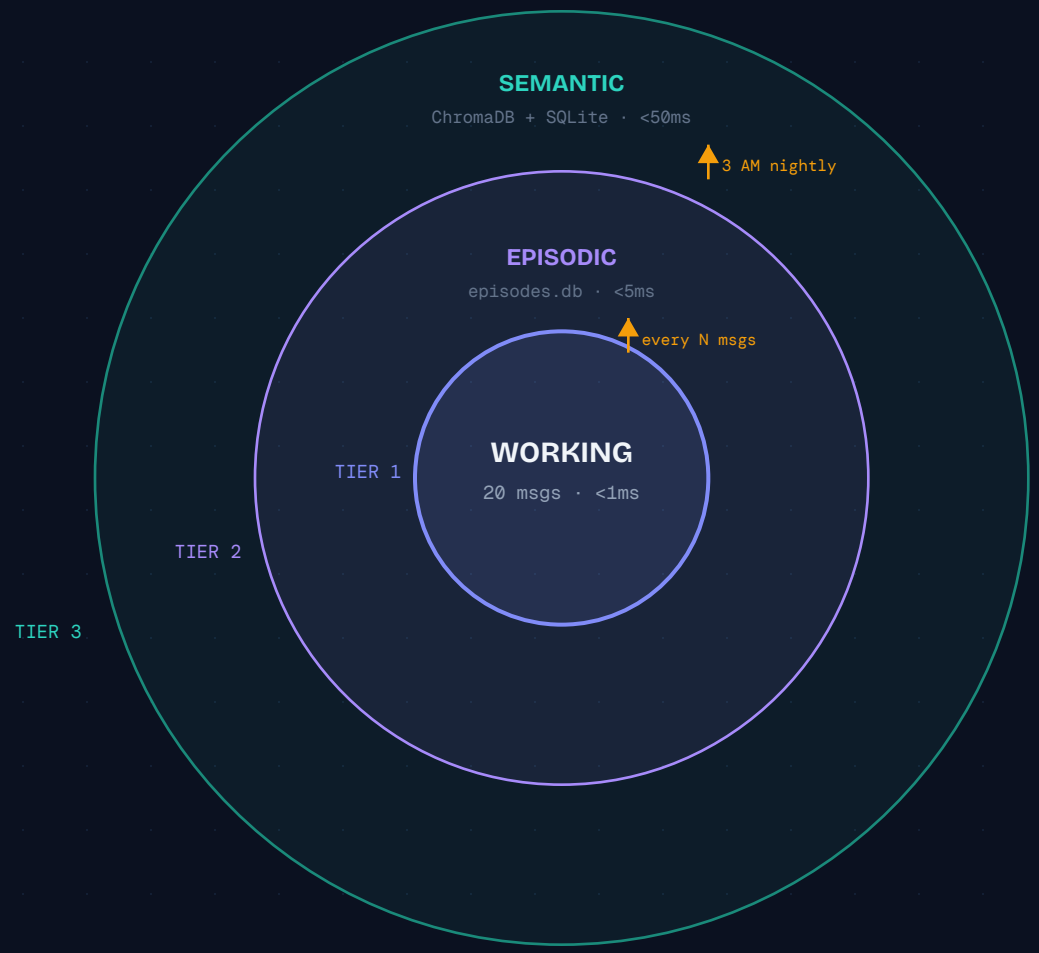
Jarvis Phase 2 — Cognitive Operating System Architecture

14 ADAPTERS · 8 SPECIALISTS · 3 MEMORY TIERS



MEMORY ARCHITECTURE

Brain-Inspired Three-Tier Hierarchy with Consolidation



CONSOLIDATION ENGINE

"Sleep Cycle" — Nightly at 3 AM

01 Replay unconsolidated episodes
02 Extract patterns via LLM
03 Merge into semantic store
04 Mark episodes consolidated
05 Prune low-value old episodes

ATTENTION GATE

Relevance-weighted prompt injection

0.35	Semantic similarity
0.20	Recency
0.15	Access frequency
0.15	Domain alignment
0.15	Surprise value

PROVENANCE CHAIN

Full audit trail for every fact

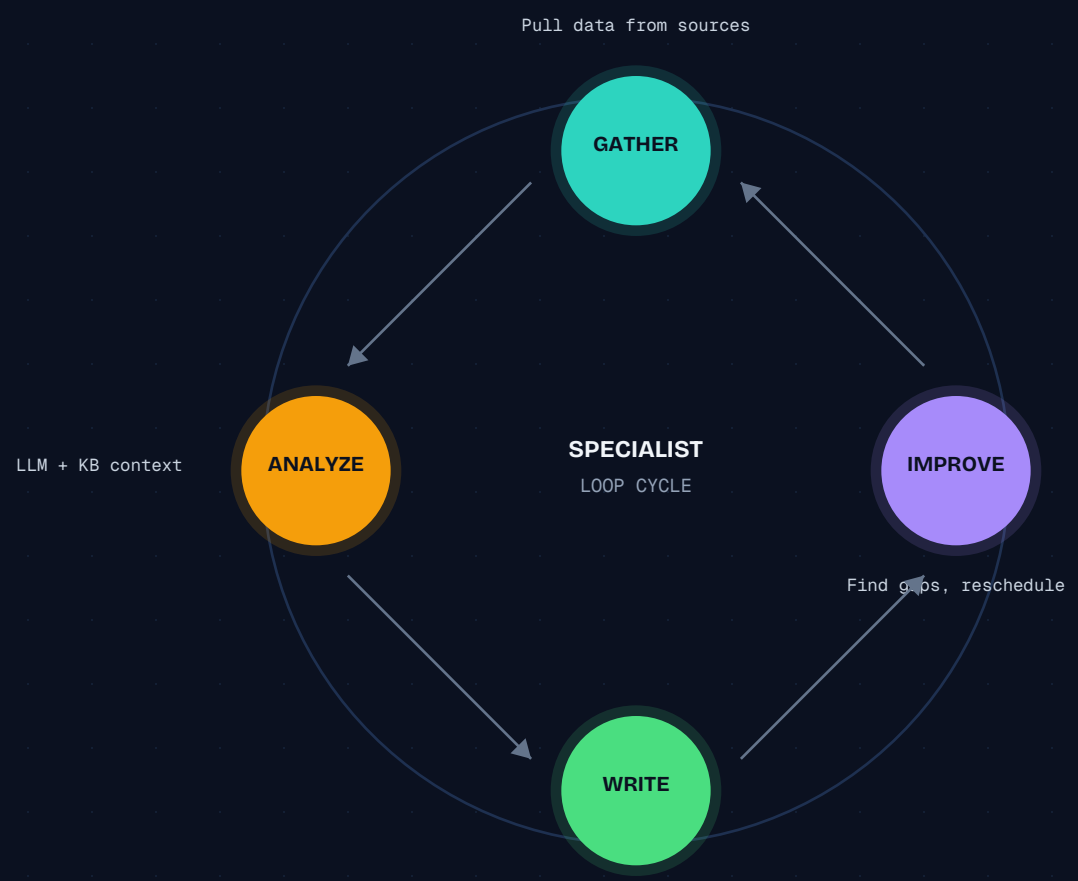
- source_type
- model_used
- prompt_hash
- confidence
- agent

ZETTELKASTEN LINKING (A-MEM, NeurIPS 2025)

Every knowledge note is atomic, richly tagged, and dynamically linked. New notes evolve existing ones.

SPECIALIST LOOP LIFECYCLE

Background AI Agents — Gather / Analyze / Write / Improve



SPECIALIST REGISTRY

- **GrocerySpec** — Every 4h
  Prices, deals, meals
- **FinanceSpec** — Daily
  Budget, spending, savings
- **CalendarSpec** — Every 2h
  Schedules, conflicts
- **HomeSpec** — Daily
  Maintenance, seasonal
- **NewsSpec** — Every 6h
  Headlines, market news
- **InvestorSpec** — Every 4h
  Portfolio, risk
- **HealthSpec** — Daily
  Appointments, wellness
- **ResearchSpec** — On-demand
  Deep dives, gap filling
- **MetaCognitive** — Daily
  Watches all specialists

CROSS-SPECIALIST COMMUNICATION

Specialists communicate through the Knowledge Lake, not directly

```
GrocerySpec writes: chicken on sale $1.99/lb at Aldi
FinanceSpec reads:  updates projected monthly spend
CalendarSpec writes: birthday party Saturday
```

HOUSEHOLD STATE

Composite state affects all specialists

- normal
- school_year
- summer
- holiday
- guests
- crunch
- budget_tight

LIBRARY OF ALEXANDRIA

Self-Building Local Knowledge System — Research Wings & Catalog

GROCERY WING

RESEARCH LIBRARIAN

- recipes
- pricing
- nutrition
- deals
- USDA data

context_engine.md
guidelines.md
resources/

FINANCE WING

RESEARCH LIBRARIAN

- budgets
- tax rules
- savings
- spending
- forecasts

context_engine.md
guidelines.md
resources/

HOME WING

RESEARCH LIBRARIAN

- repairs
- maintenance
- energy
- contractors
- seasonal

context_engine.md
guidelines.md
resources/

INVESTING WING

RESEARCH LIBRARIAN

- market data
- strategies
- risk models
- papers
- filings

context_engine.md
guidelines.md
resources/

DEV/ENG WING

RESEARCH LIBRARIAN

- papers
- repos
- frameworks
- tools
- patterns

context_engine.md
guidelines.md
resources/

SHARED CATALOG

catalog.db — Cross-referenced · Searchable · Versioned · research_queue for cross-wing requests

LIBRARIAN LIFECYCLE

SURVEY → **EVALUATE** → **CATALOG** → **CURATE**

SURVEY	EVALUATE	CATALOG	CURATE
Cast wide net	Score relevance	Extract insights	Retire outdated
APIs, papers, repos	Actionability check	Store + cross-link	Identify gaps

HISTORICAL CONTEXT ENGINE

Longitudinal records that grow richer over time

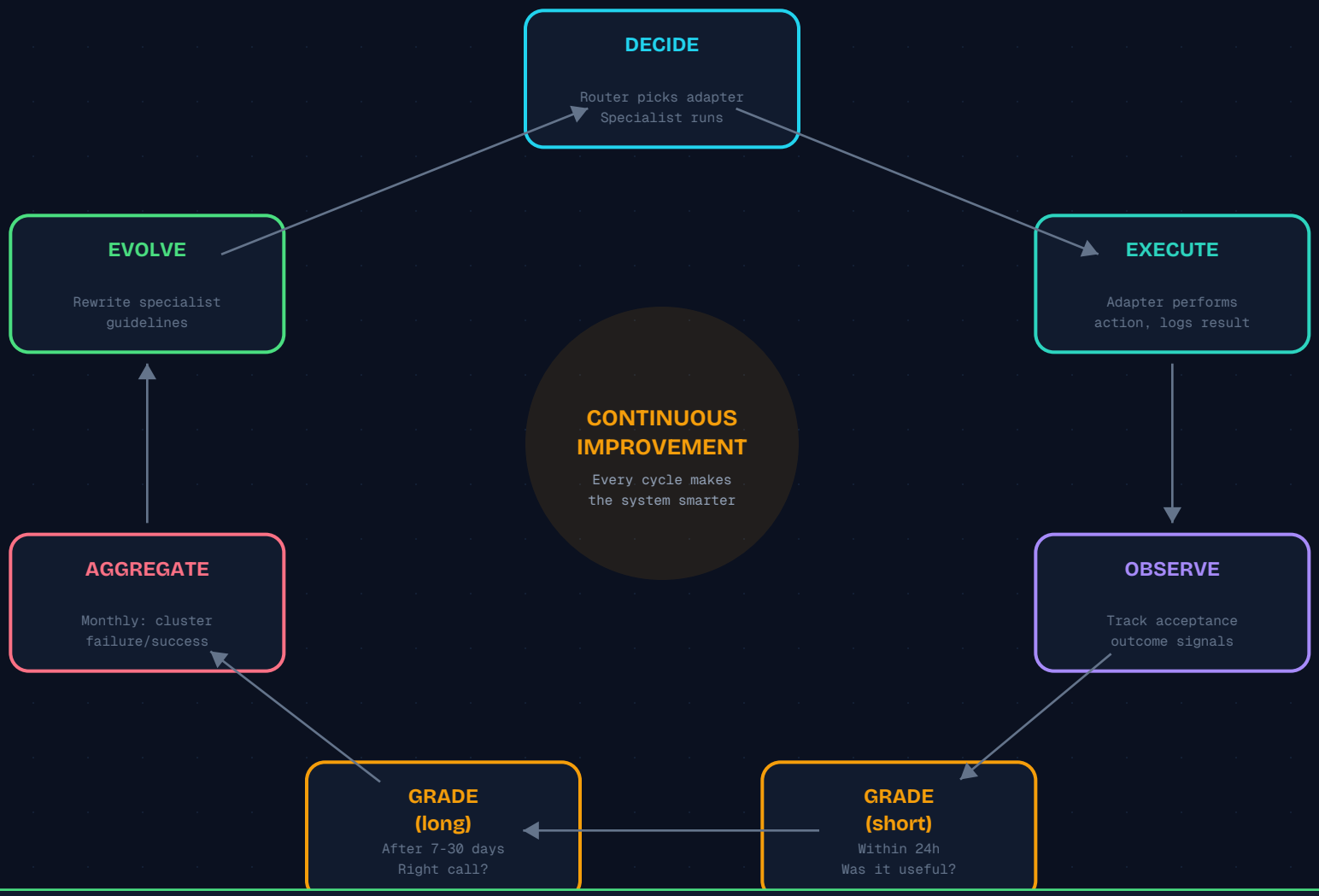
spending_history → price_history → budget_history

6 months: calibrated prediction models

12 months: seasonal pattern detection

SELF-IMPROVEMENT FEEDBACK LOOP

Decision Grading → Guideline Evolution → Context Engine Rebuild



GUIDELINE EVOLUTION EXAMPLE — GrocerySpec

v1 **3 rules**	v4 **7 rules**	v8 **10 rules**	v12 **12 rules**
Hand-written seed	+4 learned from 120 graded decisions	+3 seasonal patterns from summer data	+2 from school calendar integration

After 6 months: guidelines are battle-tested through 600+ graded decisions. The system wrote its own manual.

KNOWLEDGE ACCUMULATION ENGINES

7 Continuous World-Knowledge Databases → Model Training Pipeline

1 FINANCIAL & ECONOMIC

FRED · SEC · BLS
Yahoo · Treasury

~5M rec/yr

2 GEOPOLITICAL & EVENTS

GDELT · Congress
Reuters · UN

~2M rec/yr

3 AI RESEARCH SENTINEL

arXiv · GitHub
HuggingFace · S2

~500K rec/yr

4 LEGAL & REGULATORY

Fed Register · IRS
MN Leg · CMS

~200K rec/yr

5 HEALTH & WELLNESS

PubMed · FDA
AQI · CDC

~300K rec/yr

6 LOCAL INTELLIGENCE

Zillow · Schools
City · Crime

~100K rec/yr

7 FAMILY & LIFE QUALITY

AllTrails · Events
Parks · Parenting

~50K rec/yr

ANNUAL VOLUME

8M+
records/year

25M+
records by year 3

7
domain shards

3
storage tiers

12-18mo
to training-ready

INGESTION BUFFER

Dedup · Score · Route · Provenance

KNOWLEDGE LAKE

Hot SQLite · Warm Archive · Cold Parquet · Sharded Vectors

MODEL TRAINING PIPELINE

ShareGPT export · DPO pairs · LoRA fine-tune

Target: 12-18 months of accumulated data train household-specific models